

VAASA AP, Finland

WMO: 029110

Lat: 63.059N

Lon: 21.755E

Elev: 26

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.2	-4.7	-16.2	2.3	-11.3	-9.1	3.4	-4.9	25.3	29.9	22.6	32.4	2.2	140	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	78.7	65.0	75.2	63.2	71.8	61.5	67.2	74.8	65.1	72.4	63.2	70.1	8.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	90.8	70.3	62.4	84.2	68.8	60.4	78.5	66.7	31.6	75.0	30.0	71.9	28.6	69.8	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.7	16.5	-14.7	82.9	9.4	3.8	-21.5	85.6	-27.0	87.9	-32.2	90.0	-39.0	92.7		
(5)			-15.1	69.8	9.3	2.6	-21.8	71.7	-27.2	73.2	-32.4	74.7	-39.1	76.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	21.2	20.8	26.9	37.8	47.5	55.7	61.7	58.8	50.8	40.3	32.3	26.7	(6)
(7)		DBStd	16.30	11.95	12.17	9.05	5.91	7.26	5.48	5.28	5.45	6.07	6.87	8.57	10.52	(7)
(8)		HDD50	4566	892	817	717	368	134	11	0	3	64	304	532	724	(8)
(9)		HDD65	9118	1357	1237	1182	816	544	283	131	205	427	765	982	1189	(9)
(10)		CDD50	973	0	0	0	2	56	182	364	276	88	5	0	0	(10)
(11)		CDD65	47	0	0	0	0	1	4	29	13	0	0	0	0	(11)
(12)		CDH74	418	0	0	0	0	22	48	252	95	1	0	0	0	(12)
(13)		CDH80	54	0	0	0	0	1	6	38	9	0	0	0	0	(13)
(14)	Wind	WSAvg	7.3	7.5	7.4	7.7	7.4	8.1	7.6	6.7	6.4	6.8	7.1	7.5	7.5	(14)
(15)	Precipitation	PrecAvg	19.9	1.3	0.9	1.0	0.9	1.3	1.7	2.4	2.6	2.3	2.1	1.9	1.6	(15)
(16)		PrecMax	27.9	2.8	2.4	2.2	2.6	3.0	4.7	6.0	7.1	4.3	4.0	4.5	3.8	(16)
(17)		PrecMin	13.2	0.2	0.1	0.2	0.0	0.1	0.1	0.2	0.2	0.6	0.7	0.4	0.3	(17)
(18)		PrecStd	3.4	0.7	0.6	0.5	0.6	0.7	0.9	1.6	1.5	1.1	0.9	0.9	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	42.4	48.3	63.4	77.0	79.1	84.2	80.8	69.9	57.5	50.3	44.8	(19)
(20)			MCWB	39.1	38.4	40.7	49.9	60.0	64.0	67.2	66.5	60.3	53.0	47.9	43.8	(20)
(21)		2%	DB	38.7	39.1	43.1	57.1	71.5	74.2	80.2	76.7	66.2	54.0	47.8	42.7	(21)
(22)			MCWB	36.9	36.5	38.0	47.2	57.0	60.6	66.1	64.9	58.3	51.4	46.1	41.6	(22)
(23)		5%	DB	37.0	37.0	40.6	52.2	66.1	70.3	76.7	73.1	62.9	52.1	44.9	40.8	(23)
(24)			MCWB	35.5	35.3	36.6	44.3	53.9	58.7	64.8	63.1	56.7	49.9	43.7	39.3	(24)
(25)		10%	DB	35.3	35.2	37.7	48.5	60.9	67.6	73.2	69.6	60.7	50.2	43.1	38.8	(25)
(26)			MCWB	34.3	34.0	34.8	42.0	51.1	57.1	63.1	61.7	55.6	48.2	42.0	37.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	38.5	41.9	51.7	61.8	66.1	70.7	69.7	61.7	54.6	48.2	43.7	(27)
(28)			MCD B	40.0	40.7	47.0	59.5	72.8	75.7	79.4	76.6	67.4	56.5	49.7	44.1	(28)
(29)		2%	WB	36.8	36.5	38.7	48.0	58.4	62.9	67.9	66.5	59.7	52.2	45.8	41.4	(29)
(30)			MCD B	37.6	38.0	42.0	56.3	68.7	70.3	76.1	72.9	64.5	54.1	47.0	41.9	(30)
(31)		5%	WB	35.2	35.1	36.8	45.0	54.9	60.2	66.0	64.4	58.0	50.6	43.9	38.9	(31)
(32)			MCD B	36.1	36.1	39.6	50.9	64.4	67.8	73.5	70.7	61.8	52.3	44.7	39.8	(32)
(33)		10%	WB	33.8	33.5	35.1	42.5	51.9	58.2	64.2	62.3	56.4	48.5	42.0	36.9	(33)
(34)			MCD B	34.7	34.3	37.2	48.0	59.6	65.6	71.6	68.1	60.0	50.0	42.7	37.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	11.9	13.7	15.1	17.6	16.1	16.4	14.7	10.1	7.9	9.3	(35)	
(36)			MCD BR	8.7	9.4	13.9	23.0	26.2	23.1	22.4	21.8	18.0	11.0	8.0	7.9	(36)
(37)			MCWBR	8.0	8.0	9.8	14.4	13.8	12.1	11.2	11.7	11.4	8.7	7.3	7.3	(37)
(38)		5% WB	MCD BR	8.3	8.2	12.2	21.0	23.5	19.5	19.4	18.3	15.4	9.8	7.8	7.7	(38)
(39)			MCWBR	7.9	7.3	9.1	13.8	13.2	11.3	10.8	10.7	10.6	8.2	7.3	7.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.241	0.297	0.320	0.358	0.355	0.353	0.374	0.361	0.336	0.318	0.381	0.350	(40)	
(41)		taud	2.067	2.197	2.288	2.326	2.430	2.462	2.426	2.455	2.511	2.380	2.666	2.553	(41)	
(42)		Ebn at Noon	122	201	246	261	273	276	265	259	244	195	113	67	(42)	
(43)		Edh at Noon	10	20	27	32	31	31	31	28	22	17	9	6	(43)	
(44)	All-Sky Solar Radiation	RadAvg	47	227	705	1206	1660	1852	1696	1296	761	292	71	24	(44)	
(45)		RadStd	6	45	80	123	160	140	150	117	74	56	13	2	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page